

Dharunkumar C

Python GenAI Engineer | Generative AI & LLM Applications

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PROFESSIONAL SUMMARY

IT graduate with hands-on experience building Python-based Generative AI and machine learning applications using **LLMs, RAG, embeddings, semantic search, LangChain, LangGraph, and cloud AI services**. Experienced in developing production-oriented AI workflows for **fraud analysis, healthcare claim adjudication, document processing, retrieval, decision support, and conversational applications**. Skilled in building modular, API-driven applications using **FastAPI** and working with vector databases and enterprise data stores. Experienced in **AI evaluation, observability, monitoring, guardrails, auditability, access control, and human-in-the-loop workflows**. Strong foundation in software engineering, machine learning, computer vision, data processing, and production-grade AI application development.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

Generative AI & LLMs: Generative AI, LLMs, RAG, Prompt Engineering, Tool Calling, Semantic Search, Embeddings, Conversational AI, Context Management, AI Agents

GenAI Frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face

AI Platforms & Models: AWS Bedrock, Azure OpenAI, Llama 3, Claude, Amazon Titan Embeddings

Vector Databases & Data Stores: FAISS, OpenSearch Vector Database, PostgreSQL, DynamoDB, Redis, Azure Blob Storage, Amazon S3

Backend & APIs: FastAPI, Flask, Streamlit, REST APIs, API Design, Microservices

AI Evaluation & Observability: MLflow, RAGAS, Prometheus, Grafana, Model Evaluation, Prompt Evaluation, Monitoring

Cloud & Deployment: AWS Lambda, Amazon SQS, API Gateway, IAM, Docker, Kubernetes

AI Governance: Guardrails, Human-in-the-Loop, Access Control, Audit Logging, Traceability, Responsible AI

Machine Learning: Machine Learning, Deep Learning, Computer Vision, Image Processing, Object Detection, Image Segmentation

Tools: Git, Postman, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

WNS Global Solutions

Dec 2025 – Jun 2026

Generative AI Intern

- Contributed to **AI/ML solutions** involving data processing, validation, structured analysis, and quality-focused workflows.
- Built document ingestion and preprocessing workflows using **Python and LangChain** for enterprise AI applications.
- Implemented semantic retrieval using **FAISS** for document and business information retrieval.
- Processed claim and transaction data using **PostgreSQL**, applying validation and accuracy-checking workflows.
- Implemented AI-assisted workflows for **business information retrieval, summarization, missing-information identification, and decision support**.
- Designed and tested fraud-analysis workflows involving transaction patterns, customer information, compliance rules, and risk indicators.
- Evaluated AI responses for consistency, relevance, quality, and accuracy across business workflows.
- Maintained audit logs and monitoring workflows to support traceability, quality, and consistency.

Banking Fraud Alert & Risk Scoring AI Platform

Domain: Banking & Financial Services

- Designed and developed a **multi-agent GenAI workflow using LangGraph** for fraud alert analysis, risk scoring, AML validation, and escalation handling.
- Built **RAG pipelines** to retrieve banking policies, AML rules, fraud patterns, and compliance documents using **Amazon Titan Embeddings and OpenSearch Vector DB**.
- Integrated **AWS Bedrock Claude** to generate fraud investigation summaries, risk explanations, analyst recommendations, and suspicious activity narratives.
- Implemented **risk scoring logic** by combining transaction patterns, customer history, rule-based indicators, and LLM-generated reasoning.
- Developed **FastAPI endpoints** to expose fraud investigation services to internal banking applications.
- Implemented **audit logging, monitoring, and traceability** using DynamoDB and CloudWatch.
- Added **guardrails** to prevent unsafe recommendations, hallucinated compliance statements, and unauthorized data exposure.
- Evaluated RAG quality using **RAGAS metrics** such as faithfulness, answer relevance, context precision, and context recall.

Healthcare Claim Adjudication GenAI Assistant

Domain: Healthcare Insurance

- Designed and implemented a healthcare claim adjudication assistant using **LangChain** and **RAG** to support claim review and decision-making.
- Built **document ingestion pipelines** to process claim forms, medical bills, discharge summaries, policy documents, and adjudication guidelines.
- Generated embeddings using **Llama/OpenAI embedding models** and stored searchable claim knowledge in a **FAISS vector database**.
- Implemented retrieval logic to fetch relevant policy clauses, eligibility rules, CPT/ICD descriptions, and claim validation guidelines.
- Developed GenAI prompts to generate **claim summaries, approval/denial reasoning, missing-document alerts, and adjudication recommendations**.
- Integrated **Redis-based conversational memory** to support multi-turn claim processor interactions.
- Used **PostgreSQL** to store claim metadata, adjudication status, member details, and validation results.
- Implemented **batch processing using Celery** for large-volume claim document analysis.
- Tracked prompt versions, model outputs, and evaluation results using **MLflow**, and monitored system latency, claim processing success rate, and model response quality using **Prometheus** and **Grafana**.

ACADEMIC PROJECTS

Brain Tumor Detection | *Python, TensorFlow, Keras, VGG16, Flask, OpenCV, NumPy, Pandas, Matplotlib*

- Developed an AI-powered brain tumor detection system using the **VGG16 Convolutional Neural Network (CNN)** model for automated MRI image classification.
- Performed image preprocessing, augmentation, and feature extraction to improve model accuracy and generalization.
- Trained and evaluated the deep learning model on brain MRI datasets to classify tumor and non-tumor images, training on **7,000+ images** and achieving **97% accuracy**.
- Built a web-based application using **Flask** to enable real-time MRI image upload and prediction.
- Integrated **TensorFlow** and **Keras** frameworks for model training, inference, and performance evaluation.
- Applied data visualization techniques to analyze training accuracy, loss metrics, and prediction results.
- Improved diagnostic assistance by providing quick and accurate tumor detection predictions through an easy-to-use interface.

EDUCATION

Parul University

Bachelor of Technology (B.Tech) in Information Technology | **CGPA: 6.80**

Vadodara, India

Jul 2022 – Jun 2026

CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate

Machine Learning with Python – Great Learning Academy

Microsoft Azure AI Fundamentals (AI-900)

Google Generative AI Fundamentals

NPTEL – Computer Networks And Internet Protocol

NPTEL – Theory of Computation

NPTEL – Introduction to Internet of Things